UNIT-2					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	List out various Python List Built-in methods and write a program to demonstrate their usage.	L2	CO2	4M
	b	Write a Python program to print the even numbers from a given list.	L3	CO2	4M
2	a	Explain the steps to create a Dictionary in Python and list out Dictionary Built-in functions	L2	CO2	4M
	b	Write a Python program to perform linear search on a list.	L3	CO2	4M
3	a	What is immutable data type? Explain various string operations with suitable program.	L2	CO2	4M
	b	Write a python program to find addition of two matrices using List.	L3	CO2	4M
4	a	Is Set an ordered collection? Justify with suitable examples and also explain various operations.	L2	CO2	4M
	b	Write a python program to find trace of a matrix using Tuple.	L3	CO2	4M
5	a	Does mutability support for list, if yes explain any two methods with example?	L2	CO2	4M
	b	Distinguish between i)Lists and Tuples ii) Sets and Dictionaries.	L2	CO2	4M
6	a	Explain about Negative indexing and List Slicing in Python with a sample program.	L2	CO2	4M
	b	Writ a python program to convert various types of data into list.	L3	CO2	4M
7	a	demonstrate the following functions/methods which operates on strings in Python with suitable examples: i) len( ) ii) strip( ) iii) find( ) iv) split( ) v) index( )	L2	CO2	4M
	b	Write a python program to count number of vowels, consonants, digits in a given string.	L3	CO2	4M
8	a	Define comprehension. Classify comprehensions with suitable examples.	L2	CO2	4M
	b	Write a python script to generate multiples of n using list comprehension.	L3	CO2	4M

UNIT-1					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Discuss the int(), float(), str() and complex() type conversion functions with examples	L2	CO1	4M
	b	Write a program to print the sum of the following series 1 + 1/2 + 1/3 +. …. + 1/n	L3	CO1	4M
2	a	Describe about Membership and Identity operators with suitable examples.	L2	CO1	4M
	b	Write a Python program that reads four integers from the user, and prints them with a single print statement, without any space or newline between/after the values.	L3	CO1	4M
3	a	Explain with an example while loop, break statement and continue statement in Python.	L2	CO1	4M
	b	Write a Python program to find those numbers which are divisible by 2 and multiples of 3, between 1 and 500.	L3	CO1	4M
4	a	Define the following with the necessary examples. i. if statement ii. If-else-if statement	L2	CO1	4M
	b	Write a Python program to convert height in feet and inches to cm. [1 feet = 12 inch and 1 inch= 2.54 cm] (Sample input: 2 feet 7 inch Sample output: 78.74 cm).	L3	CO1	4M
5	a	List out and explain the key features of Python programming.	L1	CO1	4M
	b	Write a Python program that prints all the numbers from 0 to 100 except the multiples of 4 and 6.	L3	CO1	4M
6	a	Explain the execution behavior of 'for' loop statement supported by Python programming with an example program.	L2	CO1	4M
	b	Write a Python program to demonstrate the usage of Loop control statements available in Python programming.	L3	CO1	4M
7	a	Explain how an input and output function is performed in python with an example.	L2	CO1	4M
	b	Write a python program to display whether the number is Armstrong number or not?	L3	CO1	4M
8	a	List and Explain Python loop control Statements with examples	L2	CO1	4M
	b	Write a python program to display the Fibonacci sequences up to nth term where n is provided by the user.	L3	CO1	4M

ADITYA UNIVERSITY

Department of freshman Engineering

Important Questions

Course name: Programming with Python **Course code:** 2501IT42

UNIT-3					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Define function. Explain function creation and function calling in python with an example program.	L2	CO3	4M
	b	Explain about local and global variables with suitable examples	L3	CO3	4M
2	a	Explain different function arguments with suitable examples	L2	CO3	4M
	b	Write a Python function that takes two lists and returns True if they have at least one common member.	L3	CO3	4M
3	a	Explain anonymous functions with examples.	L2	CO3	4M
	b	Explain following keywords with suitable examples. i) import ii) from iii) as	L3	CO3	4M
4	a	List and explain standard modules in Python.	L2	CO3	4M
	b	Differentiate top-down and bottom-up approaches with examples	L3	CO3	4M
5	a	Explain the advantage of packages. Explain the procedure to create a package and use the same.	L1	CO3	4M
	b	Define Recursion. Explain Recursive Functions with suitable examples.	L3	CO3	4M
6	a	Explain different function categories with examples	L2	CO3	4M
	b	Explain top-down and bottom-up approaches.	L3	CO3	4M
UNIT-4					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Define Inheritance and explain types of Inheritance with examples.	L2	CO4	4M
	b	Demonstrate Polymorphism using an example Python program.	L3	CO4	4M
2	a	Demonstrate operator overloading with an example program.	L2	CO4	4M
	b	Demonstrate file operations with a suitable Python program.	L3	CO4	4M
3	a	Demonstrate the design with classes using a case study.	L2	CO4	4M
	b	Explain the functions write and write lines with examples.	L3	CO4	4M
4	a	Demonstrate the log file writing in Python program.	L2	CO4	4M

	b	Develop a program to read Config files in Python.	L3	CO4	4M
5	a	Demonstrate read, readline, and readlines with examples.	L2	CO4	4M
	b	Describe the following terms in detail and depict an example reflecting all the below i) self ii) Constructor iii) class variable iv) Instance variable	L2	CO4	4M
6	a	List and explain object-oriented concepts.	L2	CO4	4M
	b	What are the different modes of opening a file in Python? Explain the Python 'open()' built-in function.	L3	CO4	4M
7	a	Write a Python program to create a class 'student' with members {name, branch, grade}. Define appropriate member functions for reading and displaying the student information.	L3	CO4	4M
	b	Write Python code snippets to illustrate Method overriding and Method overloading concepts.	L3	CO4	4M
UNIT-5					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	List and explain any four built-in error types in Python	L2	CO5	4M
	b	Compare terminal based with GUI-based programs.	L3	CO5	4M
2	a	Define exceptions and how to handle exceptions in a Python program?	L2	CO5	4M
	b	Develop a Python program to handle division by zero exception.	L3	CO5	4M
3	a	Demonstrate GUI-based program coding with an example.	L2	CO5	4M
	b	How to create, raise and handle user defined exceptions in Python? Illustrate with a Python program.	L3	CO5	4M
4	a	List and explain clean-up actions with examples.	L2	CO5	4M
	b	Develop a Python program to implement a single try block with multiple except blocks and trace the code for its execution.	L3	CO5	4M
5	a	Is it possible to implement multiple exception blocks in Python exception handling? Justify your answer.	L2	CO5	4M
	b	Explain the purpose of 'else' and 'finally' blocks in exception handling with a Python program.	L2	CO5	4M